



The use of big data in automobile industry: Autonomous vehicles and F1

A library-based search review

MSc Business Analytics

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Declaration

Statement of authenticity:

This Project is an original and authentic piece of work by me. I have fully acknowledged and referenced all secondary sources used. It has not been presented in whole or in part for assessment elsewhere. I have read the Assessment Regulations and am fully aware of the potential consequences of any breach of them.

Signed: Mohamed Risath Riyas Ahamed

Date October 5, 2025

Abstract

This research aims to conduct a comparative analysis of the data infrastructures of autonomous vehicles and Formula 1 racing to identify the influence of Big Data in supporting real-time decisions for the AVs and performance optimization for F1, based on assessing the findings of existing literature that has been published in both domains. The study opted for interpretivist research paradigm and inductive research approach supported by library-search review considering which PICO and PRISMA frameworks were used for search strategy development, screening and final selection of the articles. A total of 6 articles were selected for this review followed by the emergence of six themes by conducting the thematic analysis by Braun and Clarke (2006).

The results of this research highlighted rapid and immediate decision making as a critical factor which directly influences the safety of autonomous vehicles and hence can ideally be less than 100 milliseconds depending on the dynamic condition of traffic on the roads and the adversity of weather and other related factors that may alter the road conditions and safety spectrum for the autonomous vehicles. Conversely, for F1, results of this research highlighted that Big Data enabled the F1 racing engineers to optimize performance of the cars through leveraging the potential of real time data analysis and simulation. The results also evidenced cloud-based simulation as a time efficient approach using which the data engineers can test the operational mechanisms of the cars and performance strategies in different environments prior to the actual race which not only aids them in identifying the anomalies, but also improve the tactics that could enhance performance during the race.

Considering the data infrastructural requirements, edge computing emerged as the dominant system for the autonomous vehicles supported by the distributed systems using which real time and rapid decisions are made within the vehicle in response to the changing dynamics on the road reducing the time for decision-making significantly especially in the critical situations which require immediate tactical and operational strategies such as roadside obstacle management and braking requirements. In contrast, in the case of F1, centralized and cloud based data systems were more appropriate supported by continuous testing of the cars in different simulation environment with a focus on improving their performance through recurrent auditing and improvisation based on the extensive streams of data generated and stored in the cloud setups that help in determining the most useful racing tactics and strategies.

The data systems for both autonomous vehicles and F1 are quite different due to their distinct focus and divergent paradigms. It is quite apparent from the findings of this study that the data infrastructure for AVs is safety-critical whereas performance improvement is the ultimate focus of the F1. Therefore, the use of edge computing as an avoidant approach is viable for AVs supporting transparent and rapid decisions and the use of cloud computing is appropriate for F1 supporting efficient and well-sought racing strategy and auditing acting as pre-requisites for car efficiency. Given that, advanced simulation environment can be useful for both autonomous vehicles and F1 racing where they can test multiple aspects related to safety, efficiency and performance of the vehicles under unique or very rare circumstances which involve high risk, simulating which may enable them to revisit their existing systems and data infrastructure to further refine the tactics as a future-ready approach.

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Chapter 1: Introduction

1. Background

The Big Data revolution, when discussed in relation to innovation, has certainly transformed the operational dynamics of industries by facilitating high fidelity, fast-paced data-driven decisions and improved performance. This transformative impact is highly evident in the domains of autonomous vehicles (AVs) and Formula 1 (F1) racing: the two systems that not only require and rely on extensive research and advanced data-supported insights but also produce fast-flowing data streams. According to Anuradha (2015), Big Data relies on five Vs such as Volume, Velocity, Value and Veracity, the viability of which is way beyond the academic abstractions enabling both AV and F1 industries to reshape their infrastructural architectures through converting the raw data into actionable insights.

For autonomous vehicles, terabytes of data are continuously produced everyday through various inertial measurement units, GNSS/INS systems, camera sensors, GPS arrays, radars, ultrasonics, LiDAR and various other high-definition sensory arrays. For example, Krentsel et al., (2024) reported that Waymo's 5th generation autonomous vehicles fleet possessed the capacity to produce more than 8GB of raw data generated every second by employing 4 radar systems, 5 LiDAR and 29 cameras. Similarly, reports have highlighted Tesla's model S and X as another largest data producing AV fleet (Gourlay and Whitelaw, 2024) where several radar and LiDAR instrumentation generate giga-bytes to terabytes of data attributed to their telemetry scale and the mileage (Fridman et al., 2019). Successful implementation of such models requires an integration of real-time and highly efficient data-producing sensory mechanisms coupled with the low-latency decision pipelines (Schwartz et al., 2018) to streamline the velocity of data ensuring precise prediction, controlling, navigation and adaptation in accordance with the continuously altering road dynamics with an ultimate focus on vehicles safety and collision avoidance. Similarly, the integrated impact of radar, maps, LiDAR and other modalities like weather conditions, traffic conditions, noise, occlusion and environmental aspects are critical determinants of Variety and Veracity (two of five Vs of Big Data), that may challenge the reliability of system (Sun et al., 2023; Parekh et al., 2022). Given that controlling and efficiently managing all these factors that may potentially degrade data fidelity is critical to determining the total value and reliability of the system (Liu and Fan, 2020).

When compared to Formula 1 racing, the need for data-intensive approach is the same; however, the challenges do vary. For example, according to the AWS, hundreds of sensors are instrumented in a single F1 car that produces more than 1.1 million telemetry data every second (AWS, 2025), for the engineers to analyze, assess the existing race strategies, scan for anomalies, identify ways to optimize them and run simulations respectively (Liu et al., 2025). The optimal decision-making process requires a thorough analysis of the historical data collected for decades to design efficient and proactive strategies to support predictive maintenance, reduce lap time, optimize performance of the cars through effective fuel and tyres management approaches to leverage the highest value, velocity, variety and veracity (Zong et al., 2025). Given that data trends from every millisecond are critical and may cost race positions for the operators. Therefore, when compared to autonomous vehicles, in the case of F1 racing, both optimized performance, safety and regulatory compliance are the key critical factors that determine the value of these cars (Shakila and Balaaji, 2025).

Despite having stark similarities in terms of relying on big data, the infrastructural requirements for both Avs and F1 are different. For instance, for autonomous vehicles, edge computing is employed which is supported by cloud-based mechanisms to ensure long-term analysis of the sensory data for evidence-informed decision making. Additionally, safety and validation are two critical components that formulate the basis of every control decision. Conversely, for F1 racing, hybrid cloud pipelines are leveraged to ensure data centralization which is further used for predictive and simulative strategies where the focus is inclined towards maximizing speed of the cars while enhancing their reliability and regulatory compliance where safety withholds a second rank. However, despite these distinct requirements, the key infrastructural issues such as sensor fusion, fault tolerance, high throughput and data heterogeneity remain the same.

Through the analysis of recent literature, it is quite evident that abundant academic and industry research has been conducted on autonomous vehicles primarily with a focus on understanding their infrastructure, algorithms, safety aspects and decision pipeline (Sun et al., 2019; Chang et al., 2019; Hu et al., 2019; Liu et al., 2019; Familsamavati et al., 2021). Similarly, for F1 data infrastructure and telemetry strategy, extensive industry, technical and limited academic sources are available that have discussed data analysis, statistical and tactical insights and race performance strategies (Sweeny, 2016; Nimmala, 2024; Msakamali, 2024; Kavitha and Kanishk, 2025). However, a comparative view of data infused infrastructural capacities of both Avs and F1 is

still an under-researched domain which hinders a clear understanding of the potential infrastructural trade-offs, beneficial practices and challenges that can facilitate an inter-domain transfer of the workable approaches that can improve data-driven efficiency and performance of both Avs and F1. Since, the telemetry data or other F1 components are usually technical reports with limited accessibility to full reports and analysis as compared to the extensive academic and non-academic research published for Avs, therefore, this cross-domain gap of comparative learning persists. Given that the viability and vitality of comparative view of computing systems of both Avs and F1, an analysis of sensor fusion pipelines as well as latency in relation to its impact on performance and reliability improvisation in both domains cannot be undermined. Thus, this dissertation bridges the identified gap by comparing how the integration of Big Data facilitates rapid decision-making and performance improvement for autonomous vehicles and F1 racing respectively, so that transferable practices across these domains can be mapped, considering this rationale, the aim, objectives and research questions for this dissertation are as follows:

1.1. Research aim

This research aims to conduct a comparative analysis of the data infrastructures of autonomous vehicles and Formular 1 racing to identify the influence of Big Data in supporting real-time decisions for the AVs and performance optimization for F1, based on assessing the findings of existing literature that has been published in both domains. Based on this aim, the objectives (RO) are as follows:

1.2. Research objectives

The research employs the following research objectives (RO):

RO1: To assess the role of Big data in supporting real time navigation, adaptation and safety-based decision making for AVs.

RO2: To understand how Big data is leveraged to optimize the performance of F1 with a focus on feedback collected from the driver, telemetry and pit-stop strategy.

RO3: To identify the unique data sources and their relative technical infrastructural characteristics required for Avs and F1.

RO4: To compare the identified infrastructural requirements and highlight similar and differentiating aspects that may facilitate inter-domain learnings.

RO5: To recommend strategies for the future with a focus of cross-domain innovation and respective developmental opportunities.

1.3. Research questions

Considering the research objectives defined in the above section (1.2), this study is based on the following research questions:

RQ1: How does Big data enable real time decision making in autonomous vehicles?

RQ2: How does Big Data facilitate performance optimization in F1 cars?

RQ3: What are the key data infrastructural requirements for AVs and F1?

RQ4: What are the similarities and differences in data infrastructural topologies for AVs and F1.

RQ5: What are the recommended strategies to support inter-domain tradeoff of infrastructural practices between AV and F1?

1.4. Rationale for library-search

There are three reasons that justify the need and vitality of library search method to cater the aim and objectives of this research:

First, the existing academic data related to AVs and F1 is quite scattered, lacking a comparative view of the useful and challenges practices in each domain. The library search facilitates a summarized yet comprehensive view of the similar and distinct features of data infrastructural requirements, practices and challenges catering to the existing research gap. This summarized view may enable industry professionals, data analysts and technical policy makers to gain useful insights from each domain that can be beneficial or help in mitigating the challenges in other domain and vice versa.

Second, since current academic literature in the domain of F1 integration to Big Data is quite limited; despite having an extensive array of technical reports and industry research reports in this

domain, therefore, library view also highlights the gaps in existing academic literature related to F1 paving a pathway for future research.

Third, this research employs a learning-oriented approach where the existing literature is analyzed for both F1 and AVs and used to map recommended actions in the future. Therefore, library search is appropriate for conducting knowledge gap analysis, summarizing the existing evidence for the best practices and challenges in each domain.

1.5. Why is this research significant?

This research is significant both academically and practically. Academically, it addressed the outlined research gap by conducting a comparative analysis of the data infrastructural sources and requirements for AV and F1, a rarely studied topic, with an ultimate focus on the potential trade-off of cross-domain lessons.

From practical perspective, the findings related to similarities and distinctive characteristics in each domain may be useful for the system designers, AV engineers and F1 performance optimization teams to map the required telemetry styles from F1 and safety-oriented mechanisms or interpretability of AVs. Additionally, these findings may also be helpful for the strategists and policy makers in AVs, aerospace and smart city development projects providing insights into leveraging the potential of the Big data.

1.6. Research structure

After setting the scope of study in this first chapter, the second chapter entitled literature review builds the theoretical foundation of this research by explaining the related theories, aspects and concepts related to Big Data and its implementation in F1 and AV in relation to the research objectives. Third chapter discusses methodology followed by the fourth chapter that presents an analysis of the selected literature for both F1 and AV. The fifth chapter discusses the findings highlighted in chapter 4 with a focus on identifying similar and differentiating aspects coupled with framing the recommended actions. The sixth chapter concludes the whole research, identifies limitations and future direction of the study.

Chapter 2

2. Literature Review

This chapter presents a review of existing concepts, frameworks, theories and literary findings related to the role of Big Data in autonomous vehicles and Formula 1 racing, where for AVs, the focus has been confined to decision making and for F1, performance optimization has been the key focus. Based on the supporting literature, this chapter also outlines gaps, presents arguments and justifies the developed research objectives and questions for this study.

2.1. Key theoretical frameworks

This section explains and assesses the key theoretical frameworks related to the Big Data infrastructure and how it supports rapid decision making and performance optimization. An in-depth understanding of these frameworks enables the researchers to grasp multiple dimensions and variables that may be appropriate to consider while discussing the possible trade-offs.

2.1.1. Data-Information knowledge-Wisdom (DIKW)

Established in 1989, Russel Acknoff emerged as one of the pioneer contributors in the field of Data analysis by presenting a classic model of transforming raw data into insightful visuals and meaningful patterns referred to as DIKW hierarchy (Fricke, 2019). The model explains the philosophical underpinnings of converting data into information by specifying its context which develops understandability while turning the information into knowledge gained by the data interpreter facilitating its practical application by making sound decisions, also termed as wisdom to understand and practically apply the gained insights (Yousaf et al., 2018; Tomita et al., 2017).

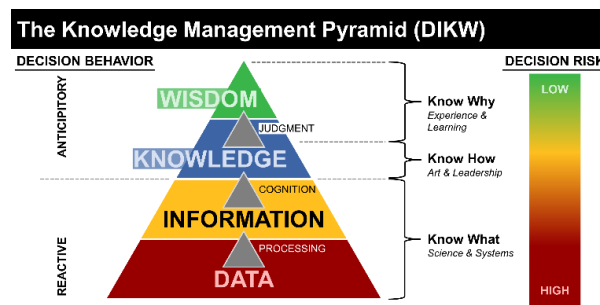


Figure 1: DIKW model

Source: D.Brown Management (2019)

Over time, this model has been modified in adherence to the dynamism of digital environment and the changes that occurred regarding the applicability and diverse yet critical role of Big data in different fields. For example, Rowley (2007) adopted an organization focus model and explained the use of Big data across various organizations describing an integration of human interaction with the data and its impact on the entire information system and interpretation design (Rowdy, 2007). Furthermore, considering the requirements and complexities of modern dynamics of Big Data environments, the model was further modified by Friecke (2019) and Baskara and Koronios (2018), who argued that the original hierarchy was rigid, static and supported linear eco-systems, which did not coincide with the modern diverse and rapid decisions making requirements and optimization needs as in AVs and F1 racing. The researchers also highlighted the need to improvise the efficiency of feedback loops by fast pacing the continuous generation, processing, storage, analysis and application of the data insights, the time lapses for which may be as minimal as a few milliseconds.

Keeping in view this, the concept of Big Data value chain was introduced that divided the process from data collection to practical application into six stages such as acquisition, storing, processing, analyzing, feedback management and implementation or action-driven decision making (Duan, 2025). This latest categorization of this hierarchal model focuses on value creation at each stage ensuring that anomalies, setbacks and bottlenecks are identified and managed at each level that may include transmission delays, analytics limitations or hardware issues, supporting a proactive approach to managing risks and enhancing the overall usefulness, accuracy and extent of implementation of data in solving real time problems (Qiu et al., 2023). For instance, autonomous vehicles employ this latest hierarchal model to generate raw sensor data, transform it into useful insights and hence make rapid driving decisions within the vehicle. Similarly, in the case of F1, telemetry systems facilitate continuous auditing, monitoring, evaluation and improvisation tactics related to improving racing performance capacity of the cars (Bi et al., 2025).

Despite these benefits and usefulness of this dynamic model, there are certain limiting aspects that may hinder the effectiveness of the model:

- First, the model does not cater all the dimensions of AVs and F1's safety and performance such as regulations, propriety, latency and safety (Rahmati, 2025).

- Second, the model assumes that the data streams undermine the value of data cleaning while assuming the collection of clean and relevant data at each stage ignoring the probability of data duplication, revision, reinterpretation and reprocessing (Bi et al., 2025).

Keeping in view these limiting aspects of DIKW model, Rossi and Hiarama (2022) proposed a new dynamic model that integrates data with people, processes, tools and models ensuring that the data value is yielded and recognized at each stage for each contributor based on design and feedback primarily which essentially supports rapid decisions in the case of AVs and performance optimization in the case of F1, transforming the classic DIKW model into a dynamic and workable framework that coincides with the continuously transforming needs and patterns of data and its use in the modern world's intelligent data environments.

2.2.2. Big data analytics (BDA) theory and organizational capability

The theory of organizational capabilities discusses the extent to which data-driven organizations adopt a progressive approach to understand, analyze, interpret and implement the emerging technologies within their business systems and infrastructural designs. When discussed within the context of BDA, the organizational capability theories explain the process of integrating the right mix of resources, cultural factors, skills and competencies to transform raw data into useful and practical insights (Shan et al., 2019; Waqar and Paracha, 2024).

The research findings by Sabharwal and Miah (2021) were a useful addition to the literature where the researchers highlighted that organizations consider various dimensions to streamline the data structures which primarily include data governance mechanisms, analytical tools, technical infrastructure and data-driven design, storing, monitoring and analysis systems. An integration of all these determinants determines the capacity of organizations to manage and interpret the complex data streams and make decisions to improve their systems, processes and operational and performance capacities respectively (Shan et al., 2019).

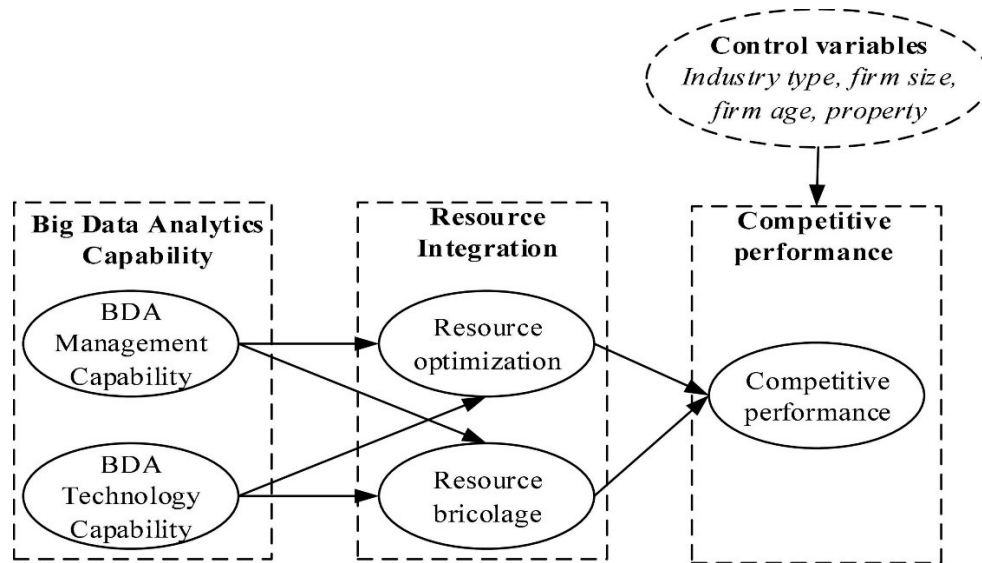


Figure 2: BDA model and its determinants

Source: Arshad et al., (2024)

For autonomous vehicles and F1 industries, BDA and organizational capabilities frameworks are highly relevant. For example, the business systems dealing with autonomous vehicles utilize extensive and continuously producing data streams to design the data management structures that not only comply with the respective regulatory requirements but also are viable to enhance the safety mechanisms of the vehicles which ultimately adds to the industry's reliability spectrum (Gupta and George, 2016; Elragal et al., 2017). Similarly, for F1, telemetry data helps in testing the performance of racing cars in different racing situations and competitive yet critical environments, which helps the system engineers test their existing capabilities, find issues and implement improved performance optimization strategies. This recurring auditing and optimization approach not only enhances the performance efficiency of the cars but also helps the companies retain competitive advantage in the industry (Gupta and George, 2016; Wamba et al., 2017). Additionally, dynamic data environments also enable the businesses to test the analytical skills of their staff and hence organize trainings accordingly ensuring that the data engineers, analysts and system designers and developers are well versed in understanding and interpreting the data structures where improved capabilities are further translated into improvised technical hardware too (Mikalef et al., 2020).

In summary, what distinguishes BDA frameworks from other theoretical aspects is that this model thinks beyond the technology and encompasses a broader scope while integrating human skills,

management aspects and governing strategies in addition to the technical aspects: all of which play a central role in amplifying the value created by the Big data in different industries. However, there are certain limiting aspects too that are attributed to BDA which are as follows:

- First, BDA theory has primarily focused on commercial and business sectors with a profound inclination towards supply chain management, retailing, banking management where the extent of sensitivity towards safety and latency is not critical when compared to the highly sensitive and high-risk environments witnessed within the autonomous vehicles and F1 industries (Sabharwal and Miah, 2021).
- Second, despite the high relevancy of BDA in data-driven environments, the above mentioned focus of the model on an integrated approach and broader view of combining technical and non-technical aspects of data management, monitoring, analysis and decisions may increase the risks of neglecting or skipping a few highly critical technical aspects such as computational delay, the extent of reliability of sensors and latency of decision making – the aspects that are reportedly not discussed within the scope of BDA; however, when analyzed within the context of AVs and F1, they are highly viable determinants (Shan et al., 2019; Waqar and Paracha, 2024).

Although there are gaps, however, the vitality of an integrated perspective for the AV manufacturers and F1 industries cannot be overlooked; given that, continuous collection and storing of data is useless if the industries lack human skills, have not developed a suitable technical infrastructure for effective data management or do not comply with the regulatory aspects. Therefore, this integrated approach advocated by the BDA models is critical to effective data maturity determining how well existing data streams have been analyzed and utilized strategically and, in a result-driven manner.

2.2.3. Real time processing and data analytics

Real time data generation, processing and analysis is another distinguished feature of Big data that supports AV and F1 mechanisms significantly. Technological frameworks like Spark, Apache Kafka and Flink support recurring streaming and analysis of data supporting an instant process of decision making where safety and performance improvements emerge as key focal points (Gupta et al., 2017). These frameworks are highly efficient in processing the data through distributed

networks and hence are central to enhancing the value of Big data in the data-intensive industries (Kreps et al., 2019; Carbone et al., 2020; Dritsas et al., 2025).

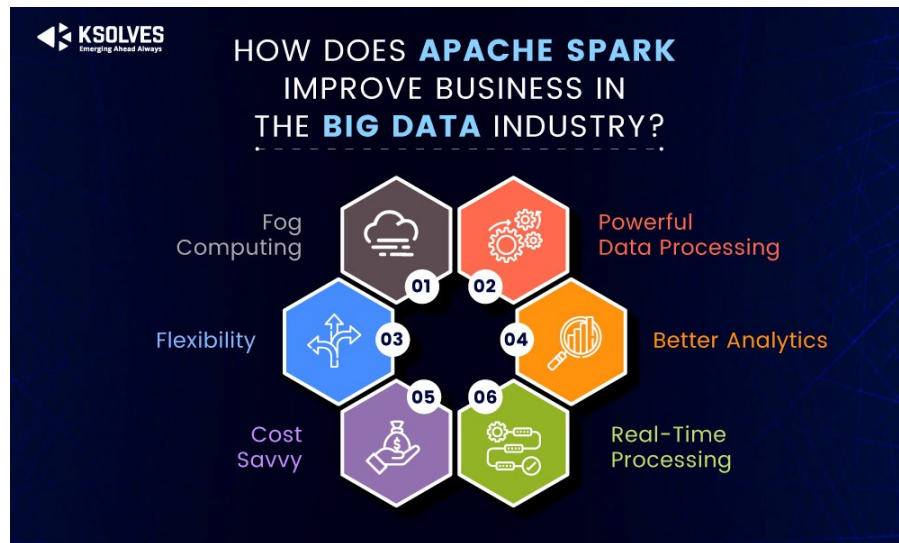


Figure 3: Big data and mechanism of performance improvement

Source: K-solves (2022)

Edge computing withholds a central position in the cloud-supported data processing infrastructures facilitating the division of computation across edge devices that are responsible for data generation such as for autonomous vehicles, these devices are termed as sensors whereas for F1 cars, centralist cloud systems are the data production and streaming sources enabling the industries to manage large quantities of data, process it and conduct large-scale yet efficiently organized and rapid analysis (Ibn-e-Kheder et al., 2021). When compared, literature has identified low-latency decision making as a distinct feature of edge computing as also evident through the studies conducted with a focus on autonomous vehicles. Conversely, for F1, cloud-supported systems facilitate long-term analysis and hence are categorized as a scalable solution to effective data management and processing (Rahmati, 2025). These data-driven infrastructural environments further support efficient decision making, speed balance and bandwidth effectiveness in the high-risk and critical environments where a delay of milliseconds can cost the businesses millions (Baidya et al., 2020).

In summary, the key benefits that are associated with real time data processing and analytics are that these aspects explicitly focus on data latency, throughput, efficiency, value creation, safety and topological aspects of the system. The systems further ensure rapid and uninterrupted travelling

and storage of data across systems where real time analytics also possess the capabilities to identify the reasons that are most likely to cause delays in data travel, streaming and analytics (Baidya et al., 2020; Rahmati et al., 2025). However, there are certainly a few limiting aspects that are needed to be acknowledged while discussing this utility and effectiveness of real time tracking frameworks in the domains of Big Data and hence are as follows:

- First, the domain agnostic nature of majority of the real time data tracking and analytics frameworks which means that these framework are specifically designed keeping in view the operational mechanisms pf the commercial, retail and banking industries and system environments rather than keeping in view the high-risk situations and critical system domains just like AVs and F1 where the chances of unforeseen situations and adversity are quite high and variable (Wang et al., 2020; Liu et al., 2021).
- Second, most of the literature already conducted in this domain has analyzed simulation environments, therefore, the challenges and operational dynamics of the systems that are deployed in real time may be quite different and hence the probability of variable performance mechanisms and safety protocols may be high and challenging for engineers with limited to no exposure to real time industrial processes (Raza et al., 2019; Liu et al., 2021). Therefore, it is apprehended that this lack of knowledge and exposure to real time industrial processes and system designs may hinder their ability to navigate the actual challenges and potential anomalies which could ultimately disrupt the efficiency and reliability of the whole system respectively (Rahmati, 2025).

2.3. The role of Big Data in AV and F1 – Supporting literature justifying the objectives

2.3.1. The role of Big data in supporting decision making in AVs

Literature has extensively identified Big data as a foundation of autonomous vehicles. Researchers have argued that the efficient operational mechanisms for the AVs require recurrent generation, processing and analysis of the real time data that is collected through sensors such as radars, cameras LiDAR or ultrasonic sensors (Sharma, 2021). However, collection of terabytes of data is useless if not analyzed and interpreted rapidly to develop actionable insights from the data that may help the data engineers to predict the potential traffic or weather-based risks, control algorithms

and make timely decisions to ensure traffic safety (Lim et al., 2019; Azmat et al., 2019; Lian et al., 2019).

Researchers have also identified sensor fusion as one of the most effective enabling techniques that facilitates integration of data from multiple sources and hence contributes to creating a highly reliable and effective environmental model (Al Najada et al., 2016; Cui et al., 2018; Raj et al., 2020). Rahmati et al., (2025) contributed to this domain and explained that integration of LiDAR with 3D cameras helps the system engineers to detect the objects accurately and precisely while making sure that the algorithms are trained to predict and manage the uncertain or unforeseen situations where the primary focus should be obstacle avoidance subject to appropriate lighting environment. Given that, this requirement of sensor fusion triggers the need for edge computing so that the computational load can be managed and hence timely decisions related to managing critical situations can be made (Cui et al., 2018; Raj et al., 2020). However, literature has also reported veracity and interpretability as keynote challenges that are subject to complexity of ML models. Although deep learning models are employed for accurate analysis; however, it gives rise to the challenges related to regulatory approvals and validation. Additionally, road conditions, lighting environment and weather adversity have also been extensively reported as considerable challenges that decrease the reliability of the entire system (Wang et al., 2023; Papa et al., 2018; Hang et al., 2020).

In summary, since there are extensive set of literature discussing the diverse role of Big data, infrastructural characteristics and data sources, the scatteredness of which hampers a summarized view of Big Data specifically for AVs with a focus on how it facilitates rapid and real time decision-making, therefore, the following research objective has been defined to bridge this gap.

Objective 1: To assess the role of Big data in supporting real time navigation, adaptation and safety-based decision making for AVs.

2.3.2. Big Data and F1 performance

When compared to AV, for formula 1 racing, the existing literature has discussed the vital role of Big Data as a pre-requisite to optimized performance of cars. Researchers have argued that data engineers and design teams working in F1 industry must process gigabytes of data during every race, which includes telemetry related to temperature of the tyres of cars in different racing

environments, aerodynamics and how it impacted the performance of cars, sub-second intervals quality as well as the engine pressure. Streamlining this data is essential to understand the underlying mechanism of analytics pipelines for F1 to see how it managed multiple aspects of performance and improvise their tactics and racing strategies accordingly. Existing literature is quite scattered where researchers have focused on different variables of performance optimization that hamper a generalized view of different dashboards and analytical measures such as pitstop analysis and overtaking etc. Therefore, intending to develop a synthesized analysis of the supportive role of Big data for F1 optimization, the following research objective has been developed:

Objective 2: To understand how Big data is leveraged to optimize the performance of F1 with a focus on feedback collected from the driver, telemetry and pit-stop strategy.

2.3.3. Data sources and infrastructure for AV and F1

Through the analysis of recent literature, it is quite evident that abundant academic and industry research has been conducted on autonomous vehicles primarily with a focus on understanding their infrastructure, algorithms, safety aspects and decision pipeline (Sun et al., 2019; Chang et al., 2019; Hu et al., 2019; Liu et al., 2019; Familsamavati et al., 2021). Similarly, for F1 data infrastructure and telemetry strategy, extensive industry, technical and limited academic sources are available that have discussed data analysis, statistical and tactical insights and race performance strategies (Sweeny, 2016; Nimmala, 2024; Msakamali, 2024; Kavitha and Kanishk, 2025). However, a comparative view of data infused infrastructural capacities of both Avs and F1 is still an under-researched domain which hinders a clear understanding of the potential infrastructural trade-offs, beneficial practices and challenges that can facilitate an inter-domain transfer of the workable approaches that can improve data-driven efficiency and performance of both Avs and F1. Since, the telemetry data or other F1 components are usually technical reports with limited accessibility to full reports and analysis as compared to the extensive academic and non-academic research published for Avs, therefore, this cross-domain gap of comparative learning persists. Given that the viability and vitality of comparative view of computing systems of both Avs and F1, an analysis of sensor fusion pipelines as well as latency in relation to its impact on performance and reliability improvisation in both domains cannot be undermined. Therefore, the

following research objectives 3, 4 and 5 were developed by facilitating a comparative view of data infrastructures supported by cross-domain lessons.

Research Objective 3: To identify the unique data sources and their relative technical infrastructural characteristics required for Avs and F1.

Based on the understandings from objective 3 about the unique data infrastructural requirements and technical aspects for AV and F1, an in-depth assessment of best and challenging practices can be framed that may support cross-domain trade-off of these practices between AVs and F1 industries serving a foundational arena for the following research objectives 4 and 5 respectively, stated as follows:

Research objective 4: To compare the identified infrastructural requirements and highlight similar and differentiating aspects that may facilitate inter-domain learnings.

Research objectives 5: To recommend strategies for the future with a focus of cross-domain innovation and respective developmental opportunities.

Chapter 3

3. Methodological design

In this chapter, the methodological design for the study has been discussed which primarily describes the process of filtering the literature and selection of 6 articles that were analyzed in this study in order to answer the five research questions developed in the first chapter in adherence to the five research objectives justified in chapter 2.

3.1. Research philosophy

In this study, interpretivist research philosophy was selected for two reasons.

First, the study aimed at conducting a comparative view of the data infrastructures of autonomous vehicles and F1 racing with a focus on understanding how each domain employs the power of Big data for fast decision making (AV) and performance optimization (F1). The findings may lead to variable tactics, situations, strategies and subjective aspects that are philosophically adherent to interpretivist research paradigms. In contrast, positivist paradigm was inappropriate because the study did not focus on abstract, objective or logical aspects, but was open to multi-dimensionality especially when focusing on comparative analysis of different determinants and aspects of Big data supported infrastructures.

Second, the study also intends to frame recommended actions for both AV and F1 industry by identifying the best practices that each domain can learn from other and implement for an optimized vehicle performance, safety and effectiveness.

3.2. Research approach

In this study, an inductive research approach was employed which facilitated the researcher to conduct a thematic analysis of the retrieved articles and extract the meaningful data insights from them that supported the pre-developed research objectives and questions. This inductive approach was also supported by abductive reasoning following which a comparative analysis was conducted leading to the identification of stark similarities and key differences in the data infrastructure of AV and F1.

3.3. Data collection approach

The first stage for data collection was identification of the key terms that were representative of the research objectives, for which PICOS was employed as follows:

PICOS table for developing search strategy		
Aspect of the model	Focus in this study	Key terms
Population	The businesses operating in autonomous vehicles industry of F1 racing cars industry	Autonomous vehicles, AV, self-driving vehicles, Formula 1, Motorsport
Intervention	Big Data technologies	Data analytics, Big Data, Machine Learning, Edge computing, Cloud computing, Artificial intelligence, Sensor fusion
Control	Comparison of AV and F1 data infrastructures	Differences, similarities, data infrastructural comparison, comparative view
Outcome	Opportunities for inter-industry transfer of best practices and lessons	Lessons learnt, best practices, value generated, challenges, hinderances, Innovation, Knowledge transfer
Study-type	Library -search	Case study, case review, Journal, research papers

Table 1: PICOS table

The data was collected from IEEE explore, Science Direct, SCOPUS and Google Scholar. The search terms defined in the PICO table were combined as a search string by using “AND” and “OR” Boolean operators so that the most relevant articles are retrieved. Furthermore, phrase searching and truncation were also activated while conducting the research from all the four data bases.

3.4. Inclusion criteria for further screening and selection

An inclusion criterion was pre-developed to ensure the selection of the most relevant articles that was applied as a key screening filter on the articles retrieved after running the initial search string within the databases. These screening aspects included the following:

- Timeline filter of 3 years was applied by ensuring the inclusion of articles that were published from January 2023 to July 2025. This timeline filter of 3 years was selected considering the dynamic and recurring nature of changes and transformative practices within the AV and F1 industry. Therefore, the intention was to conduct a comparative view of the latest and the most relevant infrastructural elements that are currently in practice.
- The studies that were published in the selected databases and were specifically focuses on data infrastructural elements for AVs and F1 industries only. All the studies that focused on other industries like aviation or the employment of Big data in commercial or retail industries were excluded in adherence to scope and objectives of the study that were defined in chapter 1.
- The studies focused on assessing the data infrastructural capacities and technologies of AVs and F1 while identifying the challenging aspects and looking for potential solutions from different industries to resolve the respective infrastructural concerns.

All the studies that were outside the scope of this inclusion criteria were excluded from the review. The initial search run retrieved a total of 1076 articles that were further screened by following different stages of PRISMA framework, an elaboration and result of each stage is described as follows:

3.5. PRISMA screening and results

PRISMA screening was a multi-stepped process which is elaborated as follows:

Stage 1: Data retrieval

At this stage, the pre-developed search string based on PICO framework was run on all the four databases leading to the retrieval of 1076 articles in total. These articles from all the data bases were imported to the citation manager as a pre-requisite to the next step of initial screening.

Stage 2: De-duplication

All the 1076 articles were screened at this stage to find all the duplicates which results in the exclusion of 354 articles. This process was conducted by running the de-duplication feature in the citation manager. Resultantly, 722 articles were selected for the next stage.

Stage 3 – Title and abstract review

At the third stage, the selected 722 articles were screened for the relevance of their research titles and abstracts to ascertain their relevance to this research's scope enabling the researcher to understand whether the articles had enough information to answer either of the research questions in this study. At this stage, a total of 523 articles were excluded, primarily due to their focus on a different industry, systematic review approach employed or a general focus on technological revolution rather than specifically on Big data. At this stage, after the exclusion of 523 articles, a total of 199 articles proceeded to the next stage.

Stage 4 – Full-text reviews

A total of 199 articles were selected for full text review, where every article was studied with a focus on deciphering their focus, methodological clarity, findings and limitations. Out of these 199 articles, 193 articles were excluded because they either lacked a critical review of the data infrastructures or had focused on a single geographical region or situation or had adopted an explanatory approach to describe various concepts of data infrastructure of AVs and F1. Resultantly a total of 6 articles were selected for this final library review that matched the scope of this study and hence had answered either of the research questions in adherence to the research objectives.

The list of selected articles included in this research is as follows:

- Chen, S., Hu, X., Zhao, J., Wang, R. and Qiao, M., (2024). A review of decision-making and planning for autonomous vehicles in intersection environments. *World Electric Vehicle Journal*, 15(3), p.99.
- Ghraizi, D., Talj, R. and Francis, C., (2023). An overview of decision-making in autonomous vehicles. *IFAC-PapersOnLine*, 56(2), pp.10971-10983.
- Li, Y., Guan, H. and Jia, X., (2024). An interpretable decision-making model for autonomous driving. *Advances in Mechanical Engineering*, 16(5), p.16878132241255455.
- Rahmati, M., (2025). Edge AI-Powered Real-Time Decision-Making for Autonomous Vehicles in Adverse Weather Conditions. *arXiv preprint arXiv:2503.09638*.

- Takács, D. and Zelei, A., (2024). Performance Optimization of a Formula Student Racing Car Using the IPG CarMaker, Part 1: Lap Time Convergence and Sensitivity Analysis. *Engineering Proceedings*, 79(1), p.86.
- Zheng, Y., Chen, Y., Tan, C., Yang, Y., Shu, C. and Chen, L., (2024). Optimization model for vehicular network data queries in edge environments. *Journal of Cloud Computing*, 13(1), p.145.

3.6. Data analysis by using thematic analysis

The selected 6 articles were further analyzed by adopting a thematic analysis approach by Braun and Clarke (2006) performed in different stages as follows:

Stage 1 – Data familiarization

At this stage, all the 6 research papers selected for this review were studied in detail and all the important information from the papers was highlighted and summarized in an excel sheet for further easy tracking of the insights from each paper.

Stage 2 – coding

Each research paper in the excel sheet was assigned a code as RP1, Rp2 and so on to RP6, as a referral code to be used while generating themes from the data.

Stage 3 – Theme development

At this stage, all the important and highlighted information was reviewed for similarities and differences discussed for AVs and F1 racing, based on which distinct classes were created.

Stage 4 and 5 – Mapping themes and naming them

Each distinct aspect identified in the stage 3 was put in one class depending on their association with the aspects discussed in the class followed by assigning them a unique name (thematic name) which ultimately resulted in the emergence of 6 themes.

Stage 6: Reporting

The themes that emerged from the study are further discussed in detail in the next chapter four, results section.

Chapter 4

4. Results and discussion

This chapter outlines the themes that emerged from the six selected articles that were analyzed in this library review.

4.1. Theme 1 – Real time decision-making and latency

4 research papers included in this review discussed latency as a dominant construct for both AVs and F1 (Rahmati et al., 2025). For example, Rahmati et al., (2025) reported effective role of edge AI in triggering safety mechanisms among AVs in response to the adversity of weather conditions where the decision-making latency was decreased from 20ms to 40ms. Similar findings were evident from the research findings of Gharaizi et al., (2023) who highlighted three categories of low-latency decision-making such as strategic, operational and tactical levels, given the variability in their temporal demands respectively. The researchers further argued that real-time and imperative decisions at operational level in AVs subject to edge AI require sub-100ms reaction times.

Conversely, in the case of F1 racing cars, latency is more concerned about the extent of competitive advantage dominating the safety imperative or immediacy of decisions: a key focal point in AVs. For instance, according to Takacs and Zelei (2024), sensitivity analysis supported by Big Data in F1 cars facilitates off-track simulations based on critical analysis and data adjustments with a variability of milliseconds. Given that, cloud platforms support real time telemetry where the time span for latency adjustments is increased from milliseconds as in offtrack simulations to second in cloud-supported environments, with the driver as being an ultimate decision-making authority (Takacs and Zelei, 2024). This stance about cloud-computing is further strengthened within the research of Zheng et al., (2024) who developed a query processing optimization model for the vehicles suggesting it as a time-efficient approach to optimize the process of data retrieval through increasing the scalability of traffic networks. The researchers further highlighted that the suggested mechanism to reduce the data retrieval time for AVs could also be relevant to F1 too having the capability to support real time telemetry data. Given that although F1 does not specifically require low latency as in the case of AVs and hence can leverage the power of Big Data in performance

optimization more vigilantly. These findings are central to the research question 1 and 2 highlighting the domain-dependency of real time data analysis and decision making.

4.2. Theme 2: Interpretability and Trust in Decision Systems

3 research papers discussed interpretability and trust as key determinants of decision system while assessing the role of Big data in AVs and F1. For example, the research findings of Li et al., (2024) evidenced the need for interpretable models arguing safety and regulatory aspects related to autonomous vehicles are supported by the black-box decision models. These models provide a rationale about vehicle behaviors in a way that can be understood by humans and hence aids in making efficient decisions which ultimately improves the trustability of the system while providing an assurance of the compliance of the system with safety standards and regulatory protocols. Similarly, the research findings by Chen et al., (2024) argued the need for the interpretability of these probabilistic models for AVs stating that more interpretabilities may facilitate the regulators, system designers and engineers to understand the data patterns more clearly and hence devise more safety-oriented tactics and strategies. Conversely, for the F1, the study findings by Takacs and Zelei (2024) argued that interpretability is a comparatively less focused aspects in the case of F1 as compared to AVs. The researchers argued that F1 primarily depends on the simulation models which require consistent auditing, monitoring, training and optimization of algorithms, where the key focus is on performance and not the extent to which models are explainable or interpretable. This stance is justified by the proclamation that high interpretability may not be required if the system design is already efficient and performance driven.

4.3. Theme 3 – Real world data vs. Simulation

5 out of six research papers discussed the big Data environment for F1 and autonomous vehicles where simulation environment emerged as a central to both domains. For example, the research findings by Takacs and Zelei (2024) highlighted that F1 testing is done within simulation environment where IGP CarMaker was categorically identified as a facilitator of sensitivity analysis that supported informed decision making about the strategies that could be used to optimize the performance of the F1 cars. However, according to the researchers, this simulation environment is a controlled dynamic where performance in different environmental conditions

prior to actual race determines the performance of vehicles during race. Similarly, it helps in determining the potential anomalies and provides enough time to the system engineers to devise optimal racing strategies.

Additionally, a similar vitality of simulation environment was evident in the research findings of Rahmati (2025) who assessed the effectiveness of edge AI in different adverse roadside conditions with a major focus on changing weather conditions. Similarly, Ghraizi et al., (2023) also identified simulation environment as a necessary mechanism for the autonomous vehicles and stated that it helps the AV engineers assume and test the potential impact of different weather conditions and how it would impact the safety of the vehicles. However, for AVs, real world challenge of data variability was identified as a critical challenge subject to considering the requirement of fast-paced decision making for AVs.

4.4. Theme 4 – Data infrastructural preferences

Considering the data infrastructural requirements, edge computing emerged as the dominant system for the autonomous vehicles supported by the distributed systems using which real time and rapid decisions are made within the vehicle in response to the changing dynamics on the road reducing the time for decision-making significantly especially in the critical situations which require immediate tactical and operational strategies such as roadside obstacle management and braking requirements (Rahmati, 2025; Ghraizi et al., 2023). In contrast, in the case of F1, centralized and cloud based data systems were more appropriate supported by continuous testing of the cars in different simulation environment with a focus on improving their performance through recurrent auditing and improvisation based on the extensive streams of data generated and stored in the cloud setups that help in determining the most useful racing tactics and strategies (fueling, tyre maintenance etc.) (Takacs and Zelei, 2024). Although both infrastructures possess unique characteristics; however, the researchers suggested to use hybrid model integrating both edge computing and cloud computing to cater diverse aspects of safety, rapid decision-making and optimized performance for both AVs and F1 highlighting the potential for F1 engineers to leverage the power of edge computing to optimize live telemetry during the actual races, whereas for AVs, cloud computing could facilitate continuous scanning and testing.

4.5. Theme 5 – Safety focused vs. performance focused approach

This theme identified two key aspects which were safety for AVs and performance for F1 cars. Researchers from the 3 studies included in this review argued that the efficient operational mechanisms for the AVs require recurrent generation, processing and analysis of the real time data that is collected through sensors such as radars, cameras LiDAR or ultrasonic sensors (Rahmati et al., 2025; Li et al., 2024; Chen et al., 2024). However, collection of terabytes of data is useless if not analyzed and interpreted rapidly to develop actionable insights from the data that may help the data engineers to predict the potential traffic or weather-based risks, control algorithms and make timely decisions to ensure traffic safety (Rahmati et al., 2025).

Researchers have also identified sensor fusion as one of the most effective enabling techniques that facilitates integration of data from multiple sources and hence contributes to creating a highly reliable and effective environmental model. Rahmati et al., (2025) contributed to this domain and explained that integration of LiDAR with 3D cameras helps the system engineers to detect the objects accurately and precisely while making sure that the algorithms are trained to predict and manage the uncertain or unforeseen situations where the primary focus should be obstacle avoidance subject to appropriate lighting environment. Given that, this requirement of sensor fusion triggers the need for edge computing so that the computational load can be managed and hence timely decisions related to managing critical situations can be made (Rahmati et al., 2025).

In contrast, in the case of F1 cars, performance is the key focus. Researchers have argued that data engineers and design teams working in F1 industry must process gigabytes of data during every race, which includes telemetry related to temperature of the tyres of cars in different racing environments, aerodynamics and how it impacted the performance of cars, sub-second intervals quality as well as the engine pressure. Streamlining this data is essential to understand the underlying mechanism of analytics pipelines for F1 to see how it managed multiple aspects of performance and improvise their tactics and racing strategies accordingly (Tackacs and Zelei, 2024; Liu et al., 2024).

4.6. Themes 6 – Opportunities for the inter-domain transfer of best practices

Four research papers that were included in this library review framed this theme by highlighting the possibilities that could spark a cross-domain transfer of knowledge between AVs and F1. Rahmati

et al., (2025) and Li et al., (2024) added to this theme significantly by highlighting the best practices for the autonomous vehicles that could be adopted by the F1 industry. Likewise, the research findings by Chen et al., (2024) and Tackacs and Zelei (2024) followed the same approach of evaluating the best and challenges infrastructural management and data analytics practices followed by F1 that could be adopted by AVs. However, this potential of bidirectional transfer of knowledge between two industries has been discussed in detail in the recommendations section of chapter 5 which also fulfills the requirements advocated by the objective 5 and research question 5 respectively.

4.7. Discussion of the themes

A total of six themes emerged from the data analysis of six research papers selected for this library-based research review. This section discusses the findings of this study in relation to the findings from other studies that have previously been published and have not adopted a comparative view. It is quite evident from the findings of the thematic analysis, when compared with the arguments from various researchers included in the literature review, the business systems dealing with autonomous vehicles utilize extensive and continuously producing data streams to design the data management structures that not only comply with the respective regulatory requirements but also are viable to enhance the safety mechanisms of the vehicles which ultimately adds to the industry's reliability spectrum (Gupta and George, 2016; Elragal et al., 2017). Similarly, for F1, telemetry data helps in testing the performance of racing cars in different racing situations and competitive yet critical environments, which helps the system engineers test their existing capabilities, find issues and implement improved performance optimization strategies. This recurring auditing and optimization approach not only enhances the performance efficiency of the cars but also helps the companies retain competitive advantage in the industry (Gupta and George, 2016; Wamba et al., 2017).

Despite having stark similarities in terms of relying on big data, the infrastructural requirements for both Avs and F1 are different. For instance, for autonomous vehicles, edge computing is employed which is supported by cloud-based mechanisms to ensure long-term analysis of the sensory data for evidence-informed decision making. Additionally, safety and validation are two critical components that formulate the basis of every control decision. Conversely, for F1 racing, hybrid cloud pipelines are leveraged to ensure data centralization which is further used for predictive and

simulative strategies where the focus is inclined towards maximizing speed of the cars while enhancing their reliability and regulatory compliance where safety withholds a second rank. However, despite these distinct requirements, the key infrastructural issues such as sensor fusion, fault tolerance, high throughput and data heterogeneity remain the same (Rahmati et al., 2025; Liu et al., 2024; Chen et al., 2024). When compared to the theoretical frameworks discussed within the literature and the supporting research evidence, Edge computing withholds a central position in the cloud-supported data processing infrastructures facilitating the division of computation across edge devices that are responsible for data generation such as for autonomous vehicles, these devices are termed as sensors whereas for F1 cars, centralist cloud systems are the data production and streaming sources enabling the industries to manage large quantities of data, process it and conduct large-scale yet efficiently organized and rapid analysis (Ibn-e-Kheder et al., 2021). When compared, literature has identified low-latency decision making as a distinct feature of edge computing as also evident through the studies conducted with a focus on autonomous vehicles. Conversely, for F1, cloud-supported systems facilitate long-term analysis and hence are categorized as a scalable solution to effective data management and processing (Rahmati, 2025). These data-driven infrastructural environments further support efficient decision making, speed balance and bandwidth effectiveness in the high-risk and critical environments where a delay of milliseconds can cost the businesses millions (Baidya et al., 2020).

Additionally, Researchers have argued that the efficient operational mechanisms for the AVs require recurrent generation, processing and analysis of the real time data that is collected through sensors such as radars, cameras LiDAR or ultrasonic sensors (Sharma, 2021). However, collection of terabytes of data is useless if not analyzed and interpreted rapidly to develop actionable insights from the data that may help the data engineers to predict the potential traffic or weather-based risks, control algorithms and make timely decisions to ensure traffic safety (Lim et al., 2019; Azmat et al., 2019; Lian et al., 2019). When compared to AV, for formula 1 racing, the existing literature has discussed the vital role of Big Data as a pre-requisite to optimized performance of cars. Researchers have argued that data engineers and design teams working in F1 industry must process gigabytes of data during every race, which includes telemetry related to temperature of the tyres of cars in different racing environments, aerodynamics and how it impacted the performance of cars, sub-second intervals quality as well as the engine pressure (Liu et al., 2024; Chen et al., 2024).

In brief, a comparative view of data infrastructure for AV and F1 highlights that the data systems for both autonomous vehicles and F1 are quite different due to their distinct focus and divergent paradigms. It is quite apparent from the findings of this study that the data infrastructure for AVs is safety-critical whereas performance improvement is the ultimate focus of the F1. Therefore, the use of edge computing as an avoidant approach is viable for AVs supporting transparent and rapid decisions (Rahmati, 2025; Ghraizi et al., 2023; Li et al., 2024) and the use of cloud computing is appropriate for F1 supporting efficient and well-sought racing strategy and auditing acting as prerequisites for car efficiency (Takacs and Zelei, 2024). Although the use of cloud computing is suggested to enhance the scalability of autonomous vehicles followed by recommending edge computing for F1 for improved tactical responsiveness of the cars (Xu et al., 2018; Zheng et al., 2024; Chen et al., 2024) however, the variability of their divergent paradigms and uniqueness of goals cannot be undermined. Given that, this suggested integration of hybrid models recommended to both domains of AV and F1 is further discussed in detail in chapter 5 (recommended actions), which explicitly identifies all the potential opportunities of cross-industry transfer of knowledge that both industries can employ for optimal safety and performance of the respective vehicles.

Chapter 5

5. Recommendations and conclusion

The last chapter presents a brief summary of the findings of this library-review based research in relation to the pre-defined objectives and respective research questions. The chapter also discusses the academic and industrial contributions of the findings of this review that may help both academicians and industry engineers and analysts working in the domains of Big Data, autonomous vehicles and F1 racing. Besides, a set of recommendations for the future in adherence to the research objective 5 has also been included within this chapter which specifically highlights the potential of inter-domain transfer of lessons related to infrastructural practices attributed to Big data in AV and F1. Lastly, the chapter outlines limitations and opportunities for future research in this field.

5.1. Concluding summary of the findings

This research aimed to understand the applicability of Big Data in the domains of autonomous vehicles and F1 racing with a focus on ascertaining how it facilitates real time decision making and performance optimization, by comparing the data infrastructures in both domains as a strategy to determine the potential and opportunities for cross-domain trade-off of lessons. Considering this aim, the research was further divided into five research objectives that are restated as follows:

RO1: To assess the role of Big data in supporting real time navigation, adaptation and safety-based decision making for AVs.

RO2: To understand how Big data is leveraged to optimize the performance of F1 with a focus on feedback collected from the driver, telemetry and pit-stop strategy.

RO3: To identify the unique data sources and their relative technical infrastructural characteristics required for Avs and F1.

RO4: To compare the identified infrastructural requirements and highlight similar and differentiating aspects that may facilitate inter-domain learnings.

RO5: To recommend strategies for the future with a focus of cross-domain innovation and respective developmental opportunities.

A brief concluding takeaway from the findings of this research in adherence to each objective is concisely synthesized as follows.

5.1.1. Objective 1 – Big Data and decision making for AVs

The results of this research highlighted rapid and immediate decision making as a critical factor which directly influences the safety of autonomous vehicles and hence can ideally be less than 100 milliseconds depending on the dynamic condition of traffic on the roads and the adversity of weather and other related factors that may alter the road conditions and safety spectrum for the autonomous vehicles (Rahmati et al., 2025; Ghraizi et al., 2023). Edge AI, as evident through the research findings of Rahmati (2025) facilitates this fast-paced ultra-low latency data-driven decision making for AVs on the road, especially considering the challenges related to unforeseen adversity of weather. The main reason for this time-efficiency aspect of Edge AI is that it allows data storage, processing and analysis within the vehicle as compared to the cloud-based storage and processing of the data (Rahmati et al., 2025; Ghraizi et al., 2023). Given that interpretability emerged as a considerable and central aspect to understand the mechanism of data processing and justification to the decisions, considering which Li et al., (2024) suggested and backed the effectiveness of transparent systems and models. The researchers apprehended that transparent models enable the data engineers to review and audit the extent of safety of vehicles considered in each decision made; however, as suggested by Doshi-Velz and Kim (2017), an equilibrium between speed and transparency is critical to determining the safety of AV infrastructures.

5.1.2. Objective 2 – Big data and performance optimization for F1

Considering the role of Big Data in F1, results of this research highlighted that Big Data enabled the F1 racing engineers to optimize performance of the cars through leveraging the potential of real time data analysis and simulation. The results also evidenced cloud-based simulation as a time efficient approach using which the data engineers have the opportunity to test the operational mechanisms of the cars and performance strategies in different environments prior to the actual race which not only aids them in identifying the anomalies, but also improve the tactics that could enhance performance during the race (Takacs and Zelei, 2024). When compared to autonomous vehicles as discussed in objective 1, performance optimization takes over latency or interpretability in the case of F1 when evaluated in integration to the role of Big data.

5.1.3. Objective 3 – Understanding the data infrastructural requirements for AV and F1

Considering the data infrastructural requirements, edge computing emerged as the dominant system for the autonomous vehicles supported by the distributed systems using which real time and rapid decisions are made within the vehicle in response to the changing dynamics on the road reducing the time for decision-making significantly especially in the critical situations which require immediate tactical and operational strategies such as roadside obstacle management and braking requirements (Rahmati, 2025; Ghraizi et al., 2023). In contrast, in the case of F1, centralized and cloud based data systems were more appropriate supported by continuous testing of the cars in different simulation environment with a focus on improving their performance through recurrent auditing and improvisation based on the extensive streams of data generated and stored in the cloud setups that help in determining the most useful racing tactics and strategies (fueling, tyre maintenance etc.) (Takacs and Zelei, 2024). Although both infrastructures possess unique characteristics; however, the researchers suggested to use hybrid model integrating both edge computing and cloud computing to cater diverse aspects of safety, rapid decision-making and optimized performance for both AVs and F1 highlighting the potential for F1 engineers to leverage the power of edge computing to optimize live telemetry during the actual races, whereas for AVs, cloud computing could facilitate continuous scanning and testing.

5.1.4. Objective 4 – Comparative view

The data systems for both autonomous vehicles and F1 are quite different due to their distinct focus and divergent paradigms. It is quite apparent from the findings of this study that the data infrastructure for AVs is safety-critical whereas performance improvement is the ultimate focus of the F1. Therefore, the use of edge computing as an avoidant approach is viable for AVs supporting transparent and rapid decisions (Rahmati, 2025; Ghraizi et al., 2023; Li et al., 2024) and the use of cloud computing is appropriate for F1 supporting efficient and well-sought racing strategy and auditing acting as pre-requisites for car efficiency (Takacs and Zelei, 2024). Although the use of cloud computing is suggested to enhance the scalability of autonomous vehicles followed by recommending edge computing for F1 for improved tactical responsiveness of the cars (Xu et al., 2018; Zheng et al., 2024; Chen et al., 2024) however, the variability of their divergent paradigms and uniqueness of goals cannot be undermined.

5.1.5. Objective 5 – Recommended actions based on possibility of inter-domain tradeoff

Considering the findings of this library-based review and similar other research articles and industry reports on Big Data and its use in F1 and AV, this section outlines a few recommended strategies that may foster inter-domain transfer of the key beneficial practices in each domain, helping each other to cater to their distinct challenges:

- ***F1 inspired telemetry dashboards for AVs:*** The developers, data engineers and system strategists working in the domain of autonomous vehicles can adopt telemetry dashboards that are used for F1 racing cars (Rahmati, 2025). Doing so may enable the AV developers to not only collect real time fleet data but also trigger continuous monitoring and auditing of the fleet performance due to the suggested centralized F1 inspired data infrastructure that will increase the chances of anomaly detection, operational challenges and challenging safety protocols, the added preventive maintenance features (Bannerjee et al., 2022) while retaining the actual focus on rapid decision making and safety aspects.
- ***Advanced simulation techniques for AVs and F1:*** Advanced simulation environment can be useful for both autonomous vehicles and F1 racing where they can test multiple aspects related to safety, efficiency and performance of the vehicles under unique or very rare circumstances which involve high risk, simulating which may enable them to revisit their existing systems and data infrastructure to further refine the tactics as a future-ready approach (Takacs and Zelei, 2024). Kalra and Paddock (2016), previously suggested the autonomous vehicles to employ simulation environment to train, test and enhance the responsive capacity of their system algorithms under adverse weather conditions, un-usual traffic conditions and multi-vehicle challenges as a proactive risk management approach; however, it has not yet been implemented. Although F1 racing already relies on simulative environments to test the performance of their cars before racing; however, they can offer training for the AV industries and new entrants in the F1 industry and guide them about the benefits and challenges they might encounter at each stage from planning to implementing and experiencing simulation (Kalra and Paddock, 2016).
- ***Foster collaborative industrial relations and knowledge sharing opportunities:*** The industrial giants in both autonomous vehicles and F1 industry can look for creating a cross-industry collaboration board or committee that may support businesses in both industries to collaborate on different data-oriented projects, learn from each other's experiences and

employ those lessons into improving the operational, strategical, analytical and sustainability aspects of their projects or products. Similarly, shared projects may also trigger intense brainstorming and innovation related to distinct aspects of telemetry integration, simulation protocols and hybrid system development; a key testing yet mandatory ground for the vehicle industries when deciphered within such a dynamic and data-rich environment (Haridasan, 2024).

- ***Need for governance protocols and cross-sector policies:*** Data governance is one of the key reported challenges for both autonomous vehicles and F1 industries where the challenges faced by F1 are more inclined towards proprietary aspects, whereas for AVs, strict regulations concerning data privacy, protection and confidentiality are a key focus (Zillner et al., 2016). Therefore, there is a need to establish standard data regulation frameworks that provide guidance to all the industries dealing with Big data (Hezam et al., 2023). This standardization will not only facilitate cross-industry learnings, but also balance regulatory compliance and innovation management, where industries may learn and employ the beneficial and result-driven strategies implemented by other businesses in other data-driven industries, learn from their experiences and hence vice versa (Ikegwu et al., 2022).

5.2. Academic and practical implications of this research

Academically, this comparative analysis of data infrastructure of AVs and F1 adds to the literature bridging the existing gap by identifying the similarities and differences into the data infrastructure for F1 and AVs and how Big data approaches can be used to facilitate inter-domain learnings. Since similar comparative studies have not been conducted previously, this research paved a foundation for more comparative literature in the future, that could be case studies, experiences from data engineers and strategists or insights from industry experts from both domains. Practically, the suggested strategies provide an insight to the industry experts such as system designers, data engineers, data analysts and industrial decision-making authorities to test run the hybrid model by integrating both edge computing and cloud computing mechanisms to introduce a new “safety-performance complied system” across both domains. Therefore, the findings of this study are viable both from academic and practical perspective.

5.3. Limitations and future direction of the study

Despite the valuable insights into the comparative view of role of Big Data in F1 and AVs, there are certain limiting aspects of this research that need to be acknowledged in order to frame a pathway for future research. First, the findings outlined in this research and themes discussed in the dissertation are derived based on the six studies related to AV and F1 that were included in this library view. Therefore, the breadth and scope of evidence is confined to the aspects that were discussed in the selected literature; given that there may be multiple technical, regulatory and proprietary aspects that may have not be included. Since, majority of the existing academic evidence is published based on simulation environments, therefore, the likelihood of variability between simulation and real time operational complexities exists. Therefore, future studies need to adopt a more practical approach by assessing the effectiveness of the data models and trained algorithms in real world industrial scenarios. Furthermore, a comparative view of multiple industries such as aviation industry and automotive technology with the AVs and F1 in terms of data sources, infrastructural requirements and data management regulations is still extinct and hence is identified as a potential future scope of this research.

5.4. Concluding statement

In summary, despite having different goals, priorities of data infrastructure and operational mechanisms majorly reflecting in safety vs performance aspects for AVs and F1, the viability and reliability of both domains on Big data is similar and unavoidable fact that creates a room for the suggested cross-domain transfer of knowledge where both industries can adopt each other's best practices and learn from the challenges they encountered while transition from safety vs. performance to introducing a hybrid data infrastructure of safety and performance across domains, for more efficient, result-driven and improvised data-driven technology eco-systems. However, it is a prolonged process that requires further academic and industrial evidence and insight that promotes selective adaptation supported by standardized regulatory and proprietary frameworks and benchmarked policies.

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Appendix 1 – Personal reflection

This section presents a critical review of my experience throughout the dissertation process starting from topic selection to final submission. At the first stage where I had to select a research topic, I always had a clarity about the domain of my research work due to my personal interest in the domain of automobiles and the dynamic nature of it being influenced by the continuously emerging Big Data technologies. However, initial research to finalize the research scope was quite challenging for me because I had previously not experienced operating the databases for articles retrievals, their analysis and gap identification. At this stage, I took guidance from the YouTube Tutorials and grasped the process of article searching. Since, it was my first experience, initially submitted proposal had a very diverse and broader scope, screening and narrowing down which to more specific aspects of Big Data and industries was another challenging activity.

After the approval of my research topic and finalizing the research objectives, drafting the first two chapters was quite easy and streamlined that I was able to complete in two weeks, however, selecting the methodological design for this library review was the next critically challenging phase which took me more than 3 weeks to read tens of articles, existing library reviews, sample papers from past students shared at the portal and consulting the YouTube videos. Resultantly, I outlined PICO, PRISMA and Thematic analysis as the most extensively used approach for data collection and analysis. As a next stage, I learnt using each one of them and implemented them at my research which was challenging yet highly learning-oriented experience that added to my existing knowledge and not only improved my research skills but also enabled me to experience the rigorous process of critical thinking and analysis of the papers. Despite being the most learning intensive phase, this process was very prolonged and required prolonged and consistent hours of working.

If I reflect on the whole journey, it was a trajectory full of learning; however, time management was a real challenge for me due to my hectic work schedules and inability to create a work life balance which hindered me from submitting this dissertation on time. Besides that, I also suffered a few health issues and familial challenges that added to the stress leaving me no choice, but to apply for an extension. In the future, I will try to create a work breakdown structure of all the activities, create a risk register and ensure that the whole project is divided into different milestones, so the issue of time management can be catered effectively.

Appendix 2 – Ethics form

EthOS Reference Number: 81988

START HERE - Basic Information

This form must be completed for all student projects.

Before you proceed

Some activities inherently involve increased risks or approval by external regulatory bodies, so a proportional ethics review is not recommended and a full ethical review may be required.

These may include:

- i. Approval from an external regulatory body (including, but not limited to: NHS (HRA), HMPPS etc.);
- ii. Misleading participants;
- iii. Research without the participants' consent;
- iv. Clinical procedures with participants;
- v. The ingestion or administration of any substance to participants by any means of delivery;
- vi. The use of novel techniques, even where apparently non-invasive, whose safety may be open to question;
- vii. The use of ionising radiation or exposure to radioactive materials;
- viii. Engaging in, witnessing, or monitoring criminal activity;
- ix. Engaging with, or accessing terrorism related materials;
- x. A requirement for security clearance to access participants, data or materials;
- xi. Physical or psychological risk to the participants or researcher;
- xii. The project activity takes place in a country outside of the UK for which there is currently an active travel warning issued by the authorities (see info button);
- xiii. Animals, animal tissue, new or existing human tissue, or biological toxins and agents;
- xiv. The sharing of participant personal data with a third party, regardless of the form under which the data is presented.

If any of these activities are fundamental to your project, please contact your supervisor to determine if a full application is required.

This form must be completed for each research project which you undertake at the University. It must be approved by your supervisor (where relevant) PRIOR to the start of any data collection.

In completing this form, please consult the University's [Research Ethics and Governance standards](#).

A1a Please confirm that you will abide by the University's Research Ethics and Governance standards in relation to this project.

- ☒ Yes
- ☐ No

A1b The University is responsible for complying with the UK General Data Protection Regulation whenever personal data is processed. Under the Data Protection Policy, all staff and students have a responsibility to comply with the regulation in their day-to-day activities and when conducting research involving personal data. The first step you can take to understand these responsibilities is to review the [Data Protection in Research guidance pages](#) and complete the University's Mandatory Data Protection Training. Data protection training is available on the [intranet data protection pages](#), and student training available through the Academic Integrity section of the [Academic Skills module in Moodle](#) (and select 'Information Governance, click here to start'). To make sure your knowledge is up to date, all staff and students must complete the training every two years where they are using personal data. If you have any issues in accessing the data protection training or have any questions about the training, please contact dataprotection@mmu.ac.uk

Have you reviewed the Data Protection guidance pages and completed the Data Protection Training in the last two years?

- ☒ Yes
☐ No

A2 Are you submitting this application as a learning experience, for a unit which already has ethical approval? (please confirm with your supervisor)

- ☒ Yes
☐ No

A2.1 Approval reference (supplied by your supervisor)

Muhammad Fayyaz

A3 Student details

Title	First Name	Surname
	Mohamed Risath	Riyas Ahamed
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A4 Supervisor

Title	First Name	Surname
mr	Muhammad	Fayyaz
Faculty	Business and Law	
Telephone	-	
Email	M.Fayyaz@mmu.ac.uk	

A5 Which Faculty is responsible for the project?

A6 Course title

A7 Project title

A8 What is the proposed start date of your project?

A9 When do you expect to complete your project?

A10 Please describe the overall aims of your project (3-4 sentences). Research questions should also be included here.

This project aims to critically examine the role of Big Data in transforming the automotive industry by analyzing its applications, benefits, and associated challenges. It focuses on how Big Data is leveraged across manufacturing, predictive maintenance, marketing, and autonomous driving through a library-based research approach using secondary sources. The key research questions include: In what ways is Big Data being applied across automotive operations? How has it contributed to predictive maintenance and customer personalization? What barriers prevent its widespread adoption, and what best practices can be drawn from leading firms?

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A11 Please describe the research activity

This research is a library-based dissertation that relies exclusively on secondary data sources. The study involves systematically reviewing academic literature, industry reports, and documented case studies related to the use of Big Data in the automotive sector. A simplified PRISMA flow diagram will be used to demonstrate the selection and screening of sources. The collected data will be analyzed thematically to identify key patterns, applications, and barriers in Big Data implementation across companies such as Tesla, BMW, and Toyota.

A12 Please provide details of the participants you intend to involve (please include information relating to the number involved and their demographics; the inclusion and exclusion criteria)

This study does not involve any human participants. As a library-based dissertation, all data will be drawn from secondary sources such as academic journals, industry reports, and publicly available case studies. Therefore, no demographic information, recruitment, or inclusion/exclusion criteria for participants apply.

A13 Please upload your project protocol

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Project Protocol	Research Protocol	Research Protocol.docx	29/07/2025	1	19.2 KB

Project Activity

B1 Are there any Health and Safety risks to the researcher and/or participants?

- ☐ Yes
- ☒ No

B2 Please select any of the following which apply to your project

- ☐ Aspects involving human participants (including, but not limited to interviews, questionnaires, images, artefacts and social media data)
- ☐ Aspects that the researcher or participants could find embarrassing or emotionally upsetting
- ☐ Aspects that include culturally sensitive issues (e.g. age, gender, ethnicity etc.)
- ☐ Aspects involving vulnerable groups (e.g. prisoners, pregnant women, children, elderly or disabled people, people experiencing mental health problems, victims of crime etc.), but does not require special approval from external bodies (NHS, security clearance, etc.)
- ☐ Project activity which will take place in a country outside of the UK
- ☒ None of the above

B2.4 Is this project being undertaken as part of a larger research study for which a Manchester Metropolitan application for ethical approval has already been granted or submitted?

- ☐ Yes
- ☒ No

Data

F1 How and where will data and documentation be stored?

All data and documentation related to this research will be securely stored on the researcher's password-protected personal computer and backed up on the university's OneDrive cloud storage associated with the official student account. Only the researcher and the academic supervisor will have access to these files. As no personal or sensitive data is being collected, data protection risks are minimal.

F2 Will you be using personal data? Personal data is anything that can be used to identify a living individual, directly or indirectly. Pseudonymised data is still personal data.

- ☐ Yes
☒ No

Insurance

F3 Does your project involve:

- ☐ Pregnant persons as participants with procedures other than blood samples being taken from them? (see info button)
- ☐ Children aged five or under with procedures other than blood samples being taken from them? (see info button)
- ☐ Activities being undertaken by the lead investigator or any other member of the study team in a country outside of the UK as indicated in the info button? If 'Yes', please refer to the 'Travel Insurance' guidance on the info button
- ☐ Working with Hepatitis, Human T-Cell Lymphotropic Virus Type iii (HTLV iii), or Lymphadenopathy Associated Virus (LAV) or the mutants, derivatives or variations thereof or Acquired Immune Deficiency Syndrome (AIDS) or any syndrome or condition of a similar kind?
- ☐ Working with Transmissible Spongiform Encephalopathy (TSE), Creutzfeldt-Jakob Disease (CJD), variant Creutzfeldt-Jakob Disease (vCJD) or new variant Creutzfeldt-Jakob Disease (nvCJD)?
- ☐ Working in hazardous areas or high risk countries? (see info button)
- ☐ Working with hazardous substances outside of a controlled environment?
- ☐ Working with persons with a history of violence, substance abuse or a criminal record?
- ☒ None of the above

Additional Information

G1 Do you have any additional information or comments which have not been covered in this form?

- ☐ Yes
☒ No

G2 Do you have any additional documentation which you want to upload?

- ☐ Yes
☒ No

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Signatures

H1 I confirm that all information in this application is accurate and true. I will not start this project until I have received Ethical Approval.

- ☒ I confirm

H2 Please notify your supervisor that this application is complete and ready to be submitted by clicking "Request" below. Do not begin your project until you have received confirmation from your supervisor - it is your responsibility to ensure that they do this.

Signed: This form was signed by Muhammad Fayyaz (M.Fayyaz@mmu.ac.uk) on 30/07/2025 11:14

H3 Have you been instructed by your supervisor to request a second signature for this application?

- ☐ Yes
☒ No

H4 By signing this application you are confirming that all details included in the form have been completed accurately and truthfully. You are also confirming that you will comply with all relevant UK data protection laws, and that that research data generated by the project will be securely archived in line with requirements specified by the University, unless specific legal, contractual, ethical or regulatory requirements apply.

Signed: This form was signed by Mohamed Risath Riyas Ahamed (MOHAMED.R.RIYAS-AHAMED@stu.mmu.ac.uk) on 29/07/2025 17:15